Network Influence Stuff

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Procrastinated forever

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1 Abstract

Approx 200 - 300 words as a very high level overview.

2 Introduction

Approx 600 - 900 words as a broad overview of the problem, challenged, direction investigated and results.

3 Literature Review

I have considered a wide range of sources so far including research papers, doctoral theses, open source datasets, articles and open source code. Each piece of information is cited or linked as a footnote where appropriate.

3.1 Models of Influence Propogation

There are three main models through which influence is typically spread across networks.

1. Independent Cascade Model (ICC) - Each node can take a state out of Active, Inactive and uninfected. Each active node has a probability p to infect each of it's uninfected neighbors. Once a node has been infected this process starts again to all of it's neighbors, the node is in the active state once it is infected [6]. Once a node has attempted to infect all of it's neighbors it becomes an inactive node. This is the simplest of the models but also mimics real life well, weather it's the spread of a new software technology or a new restaurant becoming more popular.

2. Decreasing Cascade Model (DCC) - A modification of the ICC model where every attempt of a node to be influenced decreases the probability of it being influenced or infected [7]. This could be analogous to a common experience, overreaching of advertisements or correspondence, such as marketing emails which have the opposite to intended effect due to annoyance.

3. Linear Threshold Model (LTM) - Every node v is given a probability of being turned p which is uniformly distributed across the interval $[0, 1]$. When the proportion of node v's neighbors which are infected is greater than or equal to p, v itself becomes infected [6]. This models well another common experience, a group reaching a common view in regards to something, whether it's the adoption of a new game, or even a software tool that becomes the 'industry standard' this principle is at play.

Various papers have described modifications of each of these models in a multitude of ways, multiple concepts can be spread wherein they can inhibit the spread of eachother by allowing a node to only adopt a single concept [1, 5]. Vaccination or heard immunity can be modelled in this manner [1].

There are other models which work in different manners, borrowing from biology models such as SIR for modelling diseases more closely [8]. These are somewhat able to be modelled using the cascade models however full parity is not possible for example going from infected back to uninfected is not a behavior that is considered with the cascades and would possibly have large effects on the effectiveness of certain seed selection strategies.

There are models where paths between nodes can be added or removed as part of the process [10], this can for example be used to model the evolution

of social networks, rather than studying networks which are at a fixed point in their evolution. Influence can be studied over these evolutions however this is an additional very large jump in complexity and scope which would be out of scope for this project.

3.2 Working with large networks

Some of the networks I will be studying are made up of tens of thousands of nodes with hundreds of thousands of edges, working on networks of this size is very resource intensive.

Sampling networks is a way of preserving the topological features and allows us to create strategies which generalise well to networks of a much greater size.

3.2.1 Naive Approaches

A simple natural approach could be to take a node at random from the graph, call it v, and then perform a breadth first search in layers across the graph. When a high degree node is encountered then each of it's neighbours would also be added, this will inflate the average degree of the network by tending to sample a subset with a higher mean degree [11].

Working outwards from a start node also does not preserve the distribution of shortest path distances between nodes or clustering coefficients well. For node located close to the root these values will remain quite accurate but as we move further and further out a node is less likely to appear before we have no more nodes needed leading to a lower shorter mean path length and a higher typical clustering coefficient over other methods [3].

Some papers describe BFS as adding the entire layer from which it explores, leading to oversized samples [2]. Taking BFS sample of 1000 nodes in the Enron dataset it produces on average a sample of over 12 thousand nodes which is in no way comparable to another method's 1000 node sample. This is not useful when comparing like for like network sizes and their respective topological features such as diameter which are closely dependent on overall network size.

Snowball sampling is a very simple variation on breadth first search, once the addition of the final layer would take the total number of sampled nodes over the threshold, nodes are simply added one at a time until the exact number of nodes is reached [2].

A random walk is another simple strategy for exploring a graph. Working on the basis of a standard random walk, starting from a node v there is a $\frac{1}{deg(v)}$ probability of choosing the next node to be u where u is one of the neighbours of v. intuitively RW will suffer from the same problems as BFS with finding higher degree nodes much more often than lower degrees [3], as a higher degree node has more connections that lead to it.

3.2.2 Smarter Sampling Methods

Utilising modified random walks is a standard method of achieving accurate and unbiased samples of large networks such as facebook [3]. Metropolis Hastings Random Walks and Random Walks with Delayed Acceptance (MHRW and MHDA) are two common methods used [3, 4]

(TODO MAYBE ACTUALLY EXPLAIN THIS AT ALL?)

3.3 Methods for finding seeds

There are a few methods which don't rely on a heuristic or rule of thumb to determine the seed nodes.

1. Brute Force - $\binom{n}{k}$ possible seed sets from which to choose from. This is incredibly inefficient and on the order of $O(n!)$ cascades.
2. Greedy - Run a cascade from each node and then choose the k highest influence nodes [7] giving a $1-1/e$ approximation of opt whilst being $O(n)$ cascades.
3. Cost Effective Lazy Forward (CELF) - A much faster greedy based algorithm working on only recomputing data for each node when needed [9].

The above approaches are however incredibly slow to compute for any large networks. Utilising a heuristic or approximation between a characteristic of a node and it's significance in spreading across the network can be a much more performant way of extracting similar results but at a computationally feasibly scale.

As detailed in section **??** of the specification there are a large amount of heursitics which work well as a proxy for determining how influential a node is as shown by figure **??**.

I have primarily found two main groups of heuristics, simple heuristics which consider a node and potentially it's neighbours and some more complex heuristics which consider how a node fits into the wider structure of a network.
As is quite easily seen in figure **??** and **??** these two main types are quite apparent, the rates of growth and behaviour at low seed sizes is consistent between the propogation models and network sizes with the grouping behaviour even more visible in figure **??**.

HOW DID THIS EVER GET ABOVE 45% THIS IS SO BAD TODO TALK ABOUT THE DIFFERENT HEURISTICS, WHAT THEY DO, WHY SHOULD THEY PRODUCE A HIGH INFLUENCE, WHAT THEY ACTUALLY ARE MATHEMATICALLY.

3.4 CoFIM - Community Based Methods

Talk about community based method - Requires am already partitioned network.

HOW DO YOU DEAL WITH SIZING COMMUNITIES IN NETWORKS THAT DONT HAVE A VERY STRONG COMMUNITY STRUCTURE, AT WHAT POINT DO COMMUNITIES BECOME WORTHLESS DUE TO NON COMMUNITY STRUCTURE

How do we define how strongly a network is split up into distinct communities? For a network partitioned into c communities the modularity is defined as

$$Q = \frac{1}{(2m)} \sum_{vw} \left[A_{vw} - \frac{deg_v deg_w}{(2m)} \right] \delta(c_v, c_w) = \sum_{i=1}^{c} (e_{ii} - a_i^2)$$

where e_{ij} is the proportion of edges connecting communities i and j, a_i is the fraction of ends of edges which end in community i. What is the null model in this scenario (TODO) - assuming a random graph where each vertex has a random probability of being connected to any other vertex. What we are finding via Q is how much different the proportion of edges between communities vs inside communities is in our network to the expectation of a random graph that has the same number of nodes and edges as our graph.

3.4.1 Community Detection Algorithms

Talk about top down and bottom up methods - greedy modularity and girvan newman greedy - talk about effects of chosoing a different algorithm for ranking the removal/combination of edges.

Introduced in 2006 by MEJ Newman it was found that there exists an $O(n^2 + nm)$ algorithm for community detection in networks based on the notion of maximizing modularity using a greedy method. Starting from a spectral method (vectors and matrices???) to very roughly partition the network into a set of communities. From this find vertices which will have the greatest gain in modularity for the whole graph, continuing to change the community each vertex belongs to until you cannot increase the modularity of the graph further.

Issues with partitioning a network into too many communities

3.5 Working with multiple metrics

TOPSIS - Seems to be a ranking of nodes based on placing each in a vector space based on some metrics, determining "optimal" for each metric and then choosing nodes that have the lowest distance from that optimal point in the space.

Is this good, bad? Time complexity, memory cost, size of constant factors?

Can it be easily approximated with just combining naive heuristics in a single factor. Compare to hybrid influence scores like s(v) = deg(v)*1/lcc(v)

3.6 Approximations of metrics

Randomised approximation for betweeness centrality - How good is it. how much quicker is it. Is it worthwhile to use. Is betweeness a worthwhile measure

Discouss the approxmations for local clustering - their runtimes, comparing to worst case of n cubed for high degree ndoes, how much worse do these perofrm, is it worth doing, is n cubed too slow to be useufl if more exact??

4 Problem Formulation

- A network is defined as a graph $G = (V, E)$ where $|V| = n$ and $|E| = m$
- $N(v) = \{(v, u)|u \in V\}$. The first jump neighborhood of v
- $\deg(v) = |N(v)|$
- $N_2(v) = \cup_{u \in N(v)} N(u)$ The two jump neighborhood of v
- Independent cascade model - $S = \{active, inactive, used\}$, $p \sim U[0, 1]$, $t \in [0, 1]$, A node can change from inactive to active to used only. An active node has a p chance of infecting each of it's neighbors
-

5 Proposed Algorithms

These are the ideas and directions I have explored over the time of researching this topic and am hoping will produce a measurable difference over known algorithms.

5.1 Hybrid Algorithms

From experimental testing there are clear differences in how certain methods for finding seeds perform based on the number of seeds required.

For example in the Enron dataset, a degree based approach works very well at low seed sizes however doesn't scale well with increased seed sizes. Utilising local clustering coefficient to select nodes produces an opposite pattern, at very low seed counts it performs no better than choosing a set of random nodes. Increasing the number of seeds scales very will with further seeds yielding at least a linear relationship with the final influence spread.

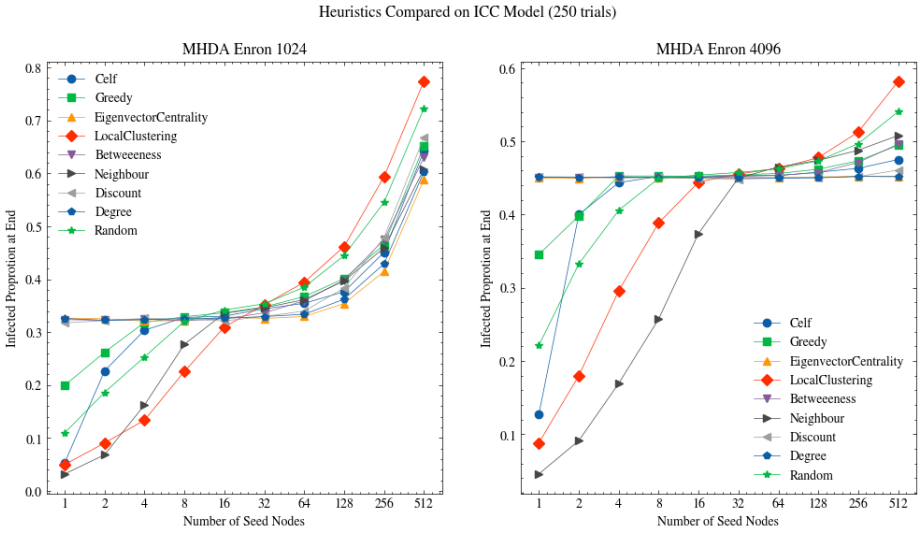


Figure 1: Heuristics Compared on an example network

The general idea this scenario is to utilise a heuristic which performs very well at low seed counts and combine it with a heuristic which performs well at higher seed counts for a significant proportion of our seeds. Based on this logic it may also be a good idea to select two or more heuristics which individually target or infect different parts of our network, where the intersection of the final spread from each individual heuristic would be as small as possible, thus maximising the final influence spread in the network more efficeintly than adding more seeds to the same approach.

Figure 1 shows the relative performance of each of the heuristics ran on a sample of the same network 250 times against a set of seed sizes expoenntially

increasing in powers of two (2) from 1 to 512. Based on this we can see that as expected local clustering coefficient performance incredibly poorly on low seed sizes but is actually the best at high seed sizes on both smaller and larger sampled networks.

1. Degree (Naive, Discount, Double Discount) and Local Clustering coefficient

2. Degree (Naive, Discount, Double Discount) and Betweenness Centrality

Why am I picking the above? - Great question (TODO). Both degree and degree discount based approaches are incredibly efficient at working with low numbers of seed nodes. Based simply on the fact that the expectation of the number of infected nodes is roughly p times the degree of the node. Discount approaches work by attempting to prevent clustering of high degree nodes being selected by removing the neighbours (or also 2 away community) of selected nodes from being selected in our seed sets.

There is a minimal overlap between the nodes chosen by clustering and betweeneess and that of degree (EXPLAIN WHY TODO).

On a large enough network with a sufficiently small sample size and a power law distribtiomn of node degrees there should not be a measurable difference in the performance of degree discount or degree approaches. We however are working with small to medium sized networks, as both a point of comparison with the standard in most papers but also to make computations take a reasonable amount of time.

Choosing nodes based on a 'score' that's made up of a combination of diffferent factors should allow for choosing nodes that will be a good overall node. If choosing some quota of nodes from one method and the remaining nodes from another the chosen set may be good individually but poor overall.

I will investigate choosing nodes that have a high degree with a low clustering, this might seem counter to the results in figure 1 where groups of nodes with high clustering perform well.

TODO WHY WILL THESE BE EFFETIVE, HOW WILL I TEST THEM, WHAT PARAMETERS DO I NEED TO CHECK, TESTING THAT THEY ARE GOOD ACROSS NETWORKS, SIZES, MODELS ETC

1. $s(v) = \deg(v) \times (1 - lcc(v))$. This will find nodes that are a combination of high degree and low clustering coefficient at the same time.

2. $s(v) = \deg(v) \times \frac{1}{lcc(v)}$. Going to the reciprocal of clustering coefficient will weight the choices more towards the lower clustering nodes than that of degree as in (1).

3. $s(v) = \deg(v) \times \frac{1}{mnd(v)}$. Choosing the nodes with the highest ratio of degree to mean neighbour degrees is quite easy to see quite similar to choosing a low clustering coefficient node because each of the nodes it's connected to will have as low an average degree as possible.

4. $s(v) = \deg(v) \times (1 - \frac{1}{max_{u \in V}(mnd(u))} mnd(v))$. In a similar vein to the relationship between (1) and (2) choose this to weight further towards the degree of a node rather than mean neighbour degree.

5. $S(k) = \deg(\log(k)) \cup lcc(k - \log(k))$. Choosing a small proportion with degree and the remaining nodes with clustering coefficient is a nice naive way of choosing these nodes. From exploratory testing there seems to be minimal overalp between the two methods in the finalised spread over a network.

6. $S(k) = \deg(\log(k)) \cup bet(k - \log(k))$. As with (5) there is a similar behaviour for betweeneess where the final spread from a set of nodes of each method has a small intersection relative to the influenced area.

Note that $s(v)$ denotes a score for each node, to choose k nodes simply choose the k nodes with the highest score. $S(k)$ denotes the set of k nodes made up of different sets.

5.2 Community Aware Algorithms

A network having a community substructure is a natural way of thinking about the internal topology of a social network. For example may have a group of friends, other people you may loosely know will have their own group of friends, finding the subset of people who make up the connection between these sets of friends intuitively will have some kind of significance in terms of spreading of ideas between the two groups.

5.2.1 Saturating or Connecting Communities

Depending upon the topology of our network the once a network is partitioned into it's constituent communities to achieve a given final spread and distribution of spread, the approach to take to achieve these goals may vary.

If there is a very prominent community then it may be enough to simply saturate that single community by targeting selection of seed nodes from only the community. This wouldn't achieve a good spread of influence globally across the network but may be the most effective method of achieving the required goals. On the other hand a network without a well defined or strong community substructure may not benefit very much from this approach, finding nodes which in turn are both good nodes within their communities but also are well connected to the other communities.

(TODO FIX THIS IS AWFUL) To measure how strongly a network contains a community structure is modularity, this is a value $Q \in [0, 1]$ which represents the proportion of edges that fall within a community compared to the expected number of edges which would fall inside a community in a random graph with the same number of edges.

Intelligently exploiting these properties of the network and it's subdivided communities may allow for more effectively maximising the spread. TODO TALK MORE ABOUT THIS. HOW DO I DETERMINE IF IT IS WORTH SATURATING A COMMUNITY RATHER THAN SPREADING SEED NODES BETWEEN COMMUNITIES

5.2.2 Exploiting Community Detection Algorithms

As described in section (TODO) there are two (2) main methods to partition the vertices of a network into distinct subsets, removing edges until there are disjoint connected components or by starting with each node in it's own community and combining them by finding the edge with the highest modularity.

When a large community splits down and it's nodes are distributed amongst the other communities, choosing nodes to be split amongst communities equally may cause a lower overall spread than having fewer larger communities.

Investigating this is a good idea as depending upon the network we may not actually get a better result by splitting into further networks (TODO).

When communities are combined into a new community (particularly in high modularity metrics) the edges which are between them can indicate that they are some of the few (very) edges which are connecting individual communities. Using the endpoints of those edges that are combined would likely act as a proxy for nodes which when influenced can spread into multiple communities. I will investigate the addition of these nodes based on the endpoints of edges that are combined into a single community using (X) heuristics, the heuristics can give a good idea of the best node to choose with the added benefit of choosign ndoes that are uniquely situated within the edges of the community structure.

TALK ABOUT HOW I DISSECTED THE COMMUNITY ALGORITHM TO FIND THE EDGES AND THEREFORE THE NODES WHICH I TRIED TO USE TO INFORM CHOOSING INFLUENTIAL NODES.

This is the algorithm for the louvain method is.

1. Consider each node to be it's own community.
2. For each node choose the neighboring community for which it will increase the modularity of the graph the most.
3.

5.3 Limited Context Algorithms

Consider a network analagous to a network graph held by google, facebook with billions of users, hundreds of connections between nodes. Performing any kind of complex algorithm within such a large network whilst not asympotically slow would be impractical in regards with speed and scale.

Selecting seeds using information obtained locally or without knowledge of the entire network, would be a useful method for working with such large networks. Metrics such as mean neighbour degree may become more valuable, inherently biased and not representative sampling methods may be reframed as a positive, a random walk containing a node itself is an indication that a node is reachable and demonstrates some baseline of influential behaviour.

NEED TO TALK MUCH MUCH MORE ABOUT THIS - NEED TO ACTUALLY IMPLEMENT IN CODE

6 Experimental Method and Setup

6.1 Datasets

1. Enron - 1 or 2 sentences - What is it, why is it good to use, how obtained, size
2. Twitter - 1 or 2 sentences
3. P2p - 1 or 2 sentences
4. Airports - 1 or 2 sentences

6.2 Sampling Methods

1. BFS - 1 or 2 sentences on behaviours and expected results
2. MHDA - 1 or 2 sentences
3. MHRW - 1 or 2 sentences

6.3 Accounting for differences in network sampling

(TODO INSERT A FIGURE OF VARIANCE BETWEEN SAMPLES AND THEIR GLOBAL METRICS)

As shown in the above table over fifteen (15) runs of a sampling algorithm on the same network the standard deviation between individual data points is quite large. With few runs and a unique sample for each one the introduction of random noise into the results may very well skew results such that a conclusion is drawn which would not be given a larger sample size.

Using an approach where for a set of trials of selecting seed nodes and finding their influence on a network and model, some small number of samples are generated and then randomly selected by each trial. This has multiple benefits over running both a single sample and sampling each time.

Working in this fashion lowers the value of the standard deviation across our runs whilst still preserving a relatively representative sample of the network.

6.4 Evaluation Methods

Measuring Final Spread, Rate of Spread, Variability due to changes in parameters, Runtime. All metrics will have a mean, standard dev computed into a tabular format to allow for very quick evaluation of statistical significance.

6.5 Baseline Behaviours

(INSERT A CUMULATIVE OR AVERAGE FIGURE FOR EACH HEURISTICS ON NETWORKS)

(INSERT A CUMULATIVE OR AVERAGE FIGURE FOR EACH SAMPLING METHOD ON NETWORKS)

TOOD talk about why the behaviour shown in the above figures is - highlight the randomness between run to run, talk about cumulative effect of choosing algorithms for a specific network being highly flawed by selected a super high performing heuristic+sample combination on one net that performs very poorly on another.

6.6 Statistical Analysis

When comparing the spread of a set of heuristics over many samples, producing a set of continous values across the range zero (0) to one (1) we can check the likelhood that the two distributions are the same. If there is found to be a statistical significance to the difference in mean and standard deviation of the final influence of two methods it gives us confidence to say that one method is statistically 'better' or 'worse' than another.

When comparing the influence produced by two methods it's important that we have a sufficient sample size to be able to determine a statistical significance. When using a t test we can use the formula

$$n = (TODO)$$

to determine the required number of samples based on the value of $d = (TODO)$, the smaller the difference in the distribution means the more samples are required to check if the variation is random based on the normal distribution of the datasets or a statistical difference.

To initially determine n we can run a pilot test of any method to determine a baseline mean and standard deviance, from this create a minimum number of runs required and round up to a nice even number. in most cases this is around 30.

7 Results and Analysis

THIS HAS ZERO RESULTS THAT ARE USEFUL OR NOVEL OR COULDNT BE THOUGHT ABOUT FOR A LITTLE BIT

8 Limitations

NOT VERY SMART - LIMITED SCOPE - POOR TIME MANAGEMENT - PROCRASTINATION - MIGHT ACTUALLY DO STUFF WITH DEADLINE IN EXACTLY A MONTH

9 Conclusion

SHORT CONCLUSION MAYBE 800-1000 WORDS SUMMARISING THE ENTIRE PROJECT, RESULTS and HOW COULD THIS BE EXPANDED FURTHER.

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